

ATFL311: AUTOMATA THEORY & FORMAL LANGUAGES
Module 3: Deterministic Finite Automata (DFA)

Comprehensive Identification & Multiple-Choice Reviewer — 41 Items

Name: _____

Date: _____ Score: _____

INSTRUCTIONS: Read each statement carefully. Identify the correct term or concept described in the blank (____). Choose the letter of the correct answer from the choices provided (a, b, c, or d) and write it on the blank provided. An exhaustive Answer Key with detailed rationales is provided at the end of this document.

PART I: IDENTIFICATION QUESTIONNAIRE (41 ITEMS)

- ____ 1. A finite state machine where for every state and for every input symbol there is exactly one unique next state, with no empty transitions allowed, is a ____ Finite Automaton.
a.) Pushdown b.) Non-Deterministic c.) Deterministic d.) Linear Bounded
- ____ 2. A fundamental rule of Deterministic Finite Automata is that empty moves (transitions without consuming an input symbol, known as ϵ -moves) are ____.
a.) unrestricted b.) strictly prohibited (not allowed) c.) optional in final states d.) required at every state
- ____ 3. Because a DFA has exactly one transition for each (state, symbol) pair, the computational path for any given input string is completely ____.
a.) random b.) branching into parallel universes c.) unique and predictable d.) probabilistic
- ____ 4. A non-accepting state in a DFA from which the machine can never escape, looping on all input symbols, is known as a dead state or ____ state.
a.) trap b.) transducer c.) initial d.) subordinate
- ____ 5. In the DFA that accepts all strings starting with '0' over $\Sigma = \{0, 1\}$ (Example 1), what happens when the first symbol read from start state A is '0'?
a.) The machine resets to state A b.) The machine enters trap state C c.) The machine transitions to accepting state B d.) The machine halts and rejects
- ____ 6. In the DFA for strings starting with '0' (Example 1), what happens when the first symbol read from start state A is '1'?
a.) The machine produces an output token b.) The machine loops at state A c.) The machine transitions to dead/trap state C d.) The machine transitions to accepting state B
- ____ 7. In the DFA for strings starting with '0' (Example 1), once the machine reaches accepting state B, what are its transitions on inputs 0 and 1?
a.) It halts immediately b.) It transitions to state C on 1 c.) It returns to start state A on 0 d.) It loops on state B for both 0 and 1
- ____ 8. How many total states are required in the minimal DFA for the language $L = \{\text{strings starting with '0'}\}$ over alphabet $\Sigma = \{0, 1\}$?
a.) 3 states (Start A, Final B, Trap C) b.) 2 states c.) 4 states d.) 5 states
- ____ 9. In DFA Example 2, the language consists of all strings over $\Sigma = \{0, 1\}$ of length exactly 2. How many strings are in this language?
a.) 16 strings b.) 4 strings ($\{00, 01, 10, 11\}$) c.) 2 strings d.) 8 strings
- ____ 10. In the DFA for strings of exact length 2 (Example 2), state C represents having consumed a string of length ____.
a.) 1 b.) 0 c.) 2 d.) 3
- ____ 11. In the DFA for strings of exact length 2 (Example 2), which state is the ONLY accepting / final state?
a.) State A b.) State B c.) State C d.) State D

- _____ 12. In the DFA for strings of exact length 2 (Example 2), what happens if a third symbol is read after reaching state C?
- a.) The machine accepts b.) The machine remains in state C c.) The machine resets to state A d.) The machine transitions to dead/trap state D
- _____ 13. In DFA Example 3, the language requires accepting all strings over $\Sigma = \{a, b\}$ that contain the substring '____'.
- a.) abab b.) aabb c.) abba d.) bbaa
- _____ 14. In the DFA for substring 'aabb', how many sequential prefix tracking states are used to reach the final state?
- a.) 4 states b.) 3 states c.) 5 states (A, B, C, D, E) d.) 6 states
- _____ 15. In the DFA for substring 'aabb', what is the transition from state C (which represents having seen 'aa') on input 'a'?
- a.) It advances to state D b.) It loops back to state C c.) It resets to start state A d.) It transitions to state B
- _____ 16. In the DFA for substring 'aabb', from state D (having seen 'aab'), what happens if the next input is 'a'?
- a.) It loops at state D b.) It returns to state B (representing prefix 'a') c.) It resets to start state A d.) It advances to final state E
- _____ 17. In the DFA for substring 'aabb', once final state E is reached, what are its outgoing transitions?
- a.) It halts immediately b.) It transitions to a trap state c.) It returns to state A on 'b' d.) It loops on both 'a' and 'b'
- _____ 18. To construct a DFA that accepts strings that DO NOT contain the substring 'aabb' (the complement language), what systematic transformation is applied to the DFA that accepts 'aabb'?
- a.) Reverse all transition arrow directions b.) Remove all self-loops c.) Invert (flip) all final and non-final states d.) Add an epsilon transition
- _____ 19. In Example 3a (the complement of 'aabb'), which states become the accepting (final) states?
- a.) {E} b.) {B, C, D} c.) {A, B, C, D} d.) {A, E}
- _____ 20. In Example 3a (the complement of 'aabb'), what state becomes the sole non-accepting (rejecting) state?
- a.) State E b.) State C c.) State B d.) State A
- _____ 21. The closure property of Regular Languages demonstrating that the complement of any regular language is also regular is proven by the ability to _____ a DFA.
- a.) minimize b.) invert (complement) c.) nondeterminize d.) tokenize
- _____ 22. In Example 4, an automaton has start state A, intermediate states B and D, accepting states C and E, and dead state X. Tracing path A → B → C on inputs 1 then 0 accepts the string '_____'.
a.) 00 b.) 11 c.) 01 d.) 10
- _____ 23. In Example 4, tracing path A → D → E on inputs 0 then 1 accepts the string '_____'.
a.) 01 b.) 00 c.) 10 d.) 11
- _____ 24. In Example 4, what happens if any additional symbol is received after reaching accepting state C or E?
- a.) The machine halts b.) The machine resets to state A c.) The machine transitions to dead state X and rejects d.) The machine loops in the accepting state
- _____ 25. The language recognized by the DFA in Example 4 is $L = \{\text{_____}\}$.
- a.) {01, 10} b.) {01, 10, 00, 11} c.) {0, 1} d.) {00, 11}
- _____ 26. In Activity 1, a DFA is constructed to accept all strings over alphabet $\Sigma = \{a, b, c\}$ of length exactly 3. How many total strings are in this language?
- a.) 9 strings b.) 27 strings (3^3) c.) 81 strings d.) 8 strings
- _____ 27. In Activity 2, for the language $L_2 = \{\text{strings that start with '0' and end with '1'}\}$, what happens to any string starting with '1'?
- a.) It loops at the start state b.) It transitions to the final state c.) It is accepted d.) It transitions immediately from the start state to a trap state
- _____ 28. In Activity 4, the given transition diagram has start state A looping on {0, 1}, transitioning on '1' to state B, and transitioning from B on {0, 1} to accepting state C. What language does this automaton recognize?

a.) All strings that end in '01' b.) All strings that start with '1' c.) All strings of length 2 d.) All strings where the second symbol from the end is '1'

_____ 29. In a DFA with alphabet $\Sigma = \{a, b, c\}$ accepting strings containing substring 'abbc' (Activity 3), how many states are required in the forward matching sequence from start to final state?

a.) 6 states b.) 5 states c.) 4 states d.) 3 states

_____ 30. In a DFA, if the start state q_0 is also an accepting state ($q_0 \in F$), the automaton accepts the _____ string.

a.) empty (ϵ) b.) infinite c.) undefined d.) dead

_____ 31. The property of a DFA that requires EVERY state to have an outgoing transition defined for EVERY symbol in alphabet Σ is known as the _____ condition.

a.) transductive b.) non-deterministic c.) completeness (totality) d.) recursive

_____ 32. In automata theory, the extended transition function that maps a state and an entire string w of symbols to a resulting state is standardly denoted as _____.

a.) Σ^* (Sigma star) b.) ϵ^* (Epsilon star) c.) Q^* (Q star) d.) δ^* (Delta star)

_____ 33. A formal representation of an automaton that uses a tabular matrix with current states as rows and input symbols as columns is a transition _____.

a.) table b.) stack c.) tree d.) tape

_____ 34. In a transition table, the start state is standardly annotated with an arrow ($\rightarrow$) and accepting states are annotated with an _____.

a.) asterisk (*) b.) ampersand (&) c.) exclamation mark (!) d.) hashtag (#)

_____ 35. In the DFA for substring 'aabb' (Example 3), if the machine is in state B (prefix 'a') and receives input 'b', why does it return to state A?

a.) Because state A is an accepting state b.) Because 'ab' cannot be part of the prefix 'aa', resetting progress c.) Because 'b' is not in the alphabet d.) Because it enters a dead state

_____ 36. In the DFA for substring 'aabb' (Example 3), what is the state trajectory when processing the input string 'baabb'?

a.) $A \rightarrow B \rightarrow A \rightarrow B \rightarrow E$ b.) $A \rightarrow C \rightarrow D \rightarrow E \rightarrow E$ c.) $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow E$ d.) $A \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$

_____ 37. In the DFA for substring 'aabb' (Example 3), does the machine accept the string 'aaabb'?

a.) Only if complemented b.) Yes, because state C loops on 'a', reaching E on 'bb' c.) No, it enters a trap state d.) No, because of too many 'a' symbols

_____ 38. In DFA design, if an automaton with 5 states is complemented by flipping final and non-final states, how many total states does the complement DFA have?

a.) 5 states b.) 6 states c.) 4 states d.) 10 states

_____ 39. For a DFA $M = (Q, \Sigma, q_0, F, \delta)$, a string w is formally accepted by M if and only if _____.

a.) $\delta^*(q_0, w) \in F$ b.) $\delta^*(q_0, w) = q_0$ c.) $\delta^*(q_0, w) \notin F$ d.) $\delta^*(q_0, w) = \phi$

_____ 40. The set of all strings accepted by a DFA M is called the language accepted by M , formally denoted as _____.

a.) $F(M)$ b.) $\Sigma(M)$ c.) $Q(M)$ d.) $L(M)$

_____ 41. Two DFAs M_1 and M_2 are defined to be equivalent if and only if they _____.

a.) accept the exact same language ($L(M_1) = L(M_2)$) b.) have the exact same number of states c.) have identical transition tables d.) share the same alphabet

COMPLETE ANSWER KEY & EXPLANATIONS

Module 3: Deterministic Finite Automata (DFA) — Reference Key with Theoretical Rationales

Item	Key	Correct Answer	Theoretical Explanation & Concept
1	C	Deterministic	[DFA Core Definition] In a Deterministic Finite Automaton (DFA), the transition function is deterministic—for each state and input symbol, there is exactly one specified next state.
2	B	strictly prohibited (not allowed)	[No Epsilon Moves in DFA] DFAs never permit empty ϵ -transitions; every transition must consume exactly one symbol from the input alphabet.
3	C	unique and predictable	[Deterministic Execution] Given an initial state and input string, a DFA follows a single, unambiguous sequence of state transitions.
4	A	trap	[Trap / Dead State Concept] A dead state (or trap state) is an absorbing non-final state where all outgoing transitions loop back to itself, ensuring rejection of the string.
5	C	The machine transitions to accepting state B	[Strings Starting with '0' Acceptance] In Example 1, on input '0' from start state A, the machine moves to state B (double circle), which is accepting.
6	C	The machine transitions to dead/trap state C	[Strings Starting with '0' Rejection] Any string beginning with '1' violates the condition and is routed immediately to trap state C, where it remains.
7	D	It loops on state B for both 0 and 1	[State B Self-Loop] Once a string is confirmed to start with '0', any subsequent symbols (0 or 1) do not change the fact that it started with '0', so state B loops on 0,1.
8	A	3 states (Start A, Final B, Trap C)	[State Count for Prefix Language] The minimal DFA requires exactly 3 states: A (start), B (accepting), and C (trap state for strings starting with 1).
9	B	4 strings ({00, 01, 10, 11})	[Length 2 Language Cardinality] Over $\Sigma = \{0, 1\}$, the strings of length 2 are 00, 01, 10, and 11, giving a cardinality of 4.
10	C	2	[Length 2 Accepting State] State A represents length 0, state B represents length 1, and state C represents length 2 (accepting).
11	C	State C	[Exact Length 2 Final State] State C is the only state designated with a double circle, representing acceptance of strings with length exactly 2.
12	D	The machine transitions to dead/trap state D	[Exceeding Length 2] Reading a third symbol exceeds the required length of 2, transitioning the automaton to trap state D, which loops on 0,1.
13	B	aabb	[Substring 'aabb' Language] Example 3 constructs a pattern matcher for the specific substring 'aabb'.

14	C	5 states (A, B, C, D, E)	[Substring Matching State Count] States correspond to prefix lengths: A (empty), B (saw 'a'), C (saw 'aa'), D (saw 'aab'), and E (saw 'aabb', accepting).
15	B	It loops back to state C	[State C Overlap Loop] If the automaton has seen 'aa' (state C) and receives another 'a', the longest relevant prefix is still 'aa', so it loops at state C.
16	B	It returns to state B (representing prefix 'a')	[State D Mismatch Fallback] After 'aab', receiving 'a' yields suffix 'a', which is represented by state B (prefix 'a').
17	D	It loops on both 'a' and 'b'	[Substring Found Absorbing State] Once the substring 'aabb' has appeared anywhere in the string, the condition is satisfied forever, so state E loops on {a, b}.
18	C	Invert (flip) all final and non-final states	[DFA Complement Algorithm] To complement a DFA, keep the exact same state set and transitions, but set $F' = Q \setminus F$ (non-accepting states become accepting, and vice-versa).
19	C	{A, B, C, D}	[Complement Accepting States] The original non-accepting states {A, B, C, D} all become accepting states (marked with double circles).
20	A	State E	[Complement Rejecting State] State E (which previously accepted 'aabb') becomes the only non-accepting state.
21	B	invert (complement)	[Closure Under Complement] Because flipping the accept/reject status of DFA states yields another valid DFA, Regular Languages are closed under complementation.
22	D	10	[Example 4 Path Trace '10'] From A, on input 1 it reaches B; from B, on input 0 it reaches accepting state C. Thus '10' is accepted.
23	A	01	[Example 4 Path Trace '01'] From A, on input 0 it reaches D; from D, on input 1 it reaches accepting state E. Thus '01' is accepted.
24	C	The machine transitions to dead state X and rejects	[Example 4 Transition to Dead State] States C and E have transitions on 0,1 pointing to dead state X, meaning only strings of length 2 ('10' and '01') are accepted.
25	A	{01, 10}	[Example 4 Language Identification] The DFA in Example 4 accepts strictly the two binary strings of length 2 with alternating digits: {01, 10}.
26	B	27 strings (3^3)	[Ternary Alphabet Length 3] With an alphabet of 3 symbols {a, b, c}, the number of strings of length 3 is $3^3 = 27$.
27	D	It transitions immediately from the start state to a trap state	[Start 0 End 1 Rejection] Since the string must start with '0', any string beginning with '1' violates the prefix requirement and is trapped.
28	D	All strings where the second symbol from the end is '1'	[Second from Last Symbol '1'] Transitioning on 1 to B followed by any single symbol {0, 1} to final state C means the string must have '1' as its penultimate (second from end) symbol.

29	B	5 states	[Substring 'abbc' State Count] Matching a 4-character substring 'abbc' requires 5 states in sequence: ϵ , 'a', 'ab', 'abb', 'abbc' (accepting).
30	A	empty (ϵ)	[Empty String Acceptance in DFA] If the initial state q_0 is a member of F , the machine accepts without reading any input symbols, meaning $\epsilon \in L(M)$.
31	C	completeness (totality)	[Completeness Condition of DFA] A DFA's transition function is total: for every state $q \in Q$ and symbol $a \in \Sigma$, $\delta(q, a)$ must be explicitly defined.
32	D	δ^* (Delta star)	[Extended Transition Function] The extended transition function $\delta^*: Q \times \Sigma^* \rightarrow Q$ computes the cumulative state transition resulting from processing an entire string.
33	A	table	[Transition Table] A transition table lists current states along the vertical axis and alphabet symbols along the horizontal axis, specifying next states in cell entries.
34	A	asterisk (*)	[Transition Table Annotations] In standard textbook notation, ' $\rightarrow$ ' marks the start state and '*' marks accepting/final states in a transition table.
35	B	Because 'ab' cannot be part of the prefix 'aa', resetting progress	[Prefix Reset Logic] The sequence 'ab' breaks the prefix 'aa'. Since 'b' cannot extend 'aa', the machine must restart matching from state A.
36	D	$A \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$	[String Trace 'baabb'] From A, on 'b' it stays at A. Then 'a' moves to B, 'a' to C, 'b' to D, and 'b' to E (accepted).
37	B	Yes, because state C loops on 'a', reaching E on 'bb'	[String Trace 'aaabb'] From A: on 'a' $\rightarrow$ B, on 'a' $\rightarrow$ C, on 'a' $\rightarrow$ C (loops), on 'b' $\rightarrow$ D, on 'b' $\rightarrow$ E (accepts).
38	A	5 states	[Complement State Preservation] Complementation changes only the subset of accepting states ($F' = Q \setminus F$); the state set Q and transition function δ remain unchanged.
39	A	$\delta^*(q_0, w) \in F$	[Formal Language Acceptance Definition] A string w is accepted by DFA M if the state reached after processing w starting from q_0 belongs to F .
40	D	$L(M)$	[Language of a Machine Notation] $L(M) = \{w \in \Sigma^* \mid \delta^*(q_0, w) \in F\}$ denotes the language recognized by automaton M .
41	A	accept the exact same language ($L(M1) = L(M2)$)	[DFA Equivalence Definition] Two automata are computationally equivalent if and only if they recognize the identical formal language.